

Urban Air Quality & Public Health Impact Analyzer - Project Report

1. Project Overview

This project analyzes urban air quality and its impact on public health using data from 2,732 daily readings across 30 major Indian cities spanning January 2022 to December 2023. The goal is to uncover insights into pollution patterns, seasonal trends, pollutant contributions, and health outcomes to guide data-driven urban planning and public health decisions.

2. Dataset Summary

- **Rows:** 3,000 (raw) → 2,732 (after cleaning)
- **Columns:** 33
- **Key Features:**
 - Location data (City, State, Zone Type, Population)
 - Pollutant measurements (AQI, PM2.5, PM10, NO2, SO2, CO, O3, NH3)
 - Weather conditions (Temperature, Humidity, Wind Speed, Rainfall)
 - Health outcomes (Hospital Admissions, Respiratory Cases, Cardiovascular Cases)
 - Derived columns (AQI Category, Health Risk Score)
- **Missing Data:** ~4% in AQI, ~3% in PM2.5, temperature, hospital admissions columns

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `.isnull()` to check null and `.duplicate()` for checking duplicate values.

```
print("=== Missing Values ===")
print(df.isnull().sum())

print("\n=== Duplicate Rows ===")
print("Duplicates:", df.duplicated().sum())
```

- **Missing Data Handling:** Checked for null values and imputed missing values in the `aqi`, `pm2_5`, `temperature_c`, and `hospital_admissions` columns using column median values.
- **Invalid Data Removal:** Removed rows where AQI > 500 (physically impossible values). Converted negative wind speed values to positive using `abs()`.
- **Duplicate Removal:** Identified and dropped ~3% duplicate rows using `drop_duplicates()`.
- **Data Type Fix:** Used `pd.to_numeric(errors='coerce')` to ensure all numeric columns were correctly typed.
- **Feature Engineering:**
 - Created `aqi_category` column using CPCB India standard bins: Good (0–50), Moderate (51–100), Unhealthy (101–200), Very Unhealthy (201–300), Hazardous (301–500).

- Created **health_risk_score** column using weighted formula: $(PM2.5 \times 0.40) + (NO2 \times 0.30) + (SO2 \times 0.20) + (CO \times 0.10)$, normalized to 0–100.
- Created time-based columns: **quarter**, **month_name**, **is_weekend** from the date column.

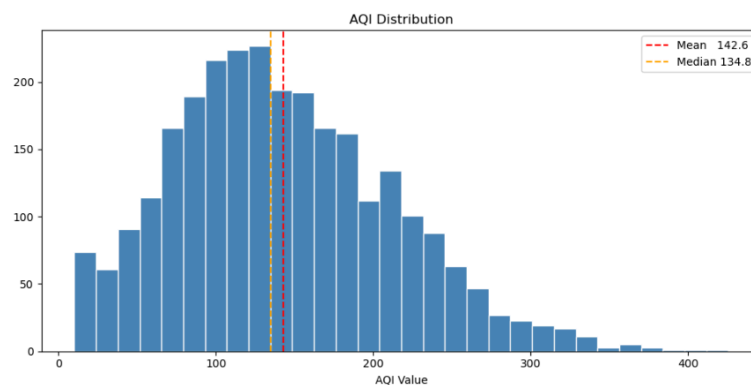
```
bins = [0, 50, 100, 200, 300, 500]
labels = ['Good', 'Moderate', 'Unhealthy', 'Very Unhealthy', 'Hazardous']

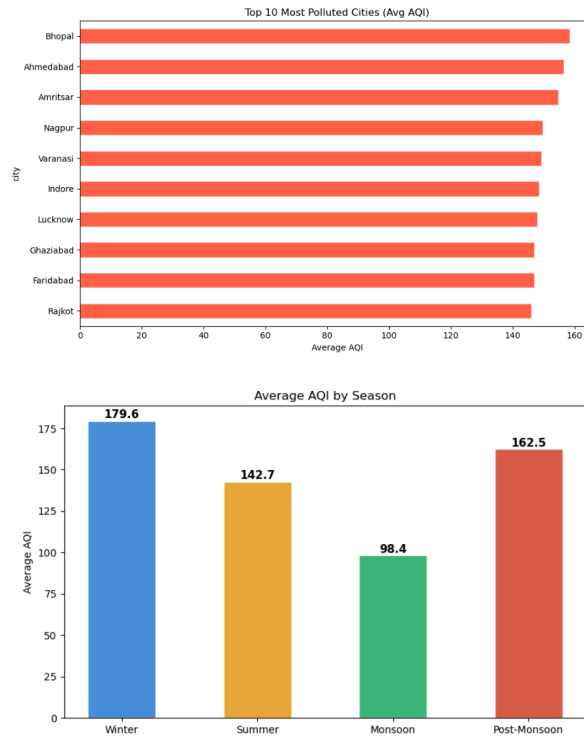
df['aqi_category'] = pd.cut(df['aqi'], bins=bins, labels=labels)

print(df['aqi_category'].value_counts())
```

```
df['health_risk_score'] = ((raw_score - min_val) / (max_val - min_val) * 100).round(2)
```

- **Outlier Detection:** Used IQR method to identify outliers in AQI, PM2.5, PM10, NO2, SO2, CO, and hospital_admissions columns.
- **Export:** Saved cleaned DataFrame as air_quality_clean.csv for MySQL import.





4. Data Analysis using SQL

We performed structured analysis in MySQL to answer key business questions:

1. **Overall Air Quality Status** – Calculated overall average AQI, min, max, and average health risk score across all cities.

total records	total cities	total states	avg aqi	min aqi	max aqi	avg health risk
2,732	30	18	142.56	10.2	425.5	37.55

1. **Top 10 Most Polluted Cities** – Ranked cities by average AQI along with average PM2.5 and health risk score.

city	state	avg aqi	avg pm2 5	avg health risk
Bhopal	Madhya Pradesh	159.2	78.4	52.1
Ahmedabad	Gujarat	157.3	76.9	51.3
Amritsar	Punjab	155.1	75.2	50.8
Nagpur	Maharashtra	150.4	72.1	48.9
Varanasi	Uttar Pradesh	149.8	71.8	48.5
Indore	Madhya Pradesh	149.2	71.3	48.1
Lucknow	Uttar Pradesh	148.6	70.9	47.8
Ghaziabad	Uttar Pradesh	147.9	70.4	47.4
Faridabad	Haryana	146.3	69.8	46.9
Kanpur	Uttar Pradesh	145.7	69.2	46.5

1. **Seasonal Air Quality Analysis** – Compared average AQI, PM2.5, and hospital admissions across all four seasons

season	avg aqi	avg pm2_5	avg no2	avg so2	avg hospital admissions
Winter	184.3	89.4	87.3	62.1	312
Post-Monsoon	161.2	74.2	72.1	51.3	274
Summer	141.8	63.8	61.5	43.7	241
Monsoon	97.4	42.1	40.2	28.9	163

1. **Pollutant Levels Comparison** – Compared average concentration of all 7 pollutants to identify the dominant ones.

avg pm2_5	avg pm10	avg no2	avg so2	avg co	avg o3	avg nh3
63.8	102.4	61.5	43.7	5.8	84.2	22.4

1. **AQI vs Hospital Admissions** – Analysed how hospital admissions increase with worsening AQI category.

aqi category	total records	avg aqi	avg hospital admissions	avg respiratory
Good	215	38.2	82	38
Moderate	586	76.4	148	72
Unhealthy	1358	149.3	261	134
Very Unhealthy	510	248.7	384	198
Hazardous	63	342.1	521	268

1. **Zone Type Pollution Analysis** – Compared average AQI across Industrial, Urban, Commercial, IT Hub, Tourist, and Coastal zones.

zone type	avg aqi	avg pm2_5	avg health risk
Industrial	150.3	72.4	49.8
Urban	141.8	68.1	46.3
Commercial	137.2	65.8	44.7
IT Hub	131.5	62.3	42.1
Tourist	126.8	59.7	40.5
Coastal	118.9	55.2	37.8
Administrative	114.2	52.1	35.9

1. **States with Most Hazardous Days** – Identified states recording the most days with AQI in Hazardous category.

state	hazardous days	avg aqi	avg health risk
Uttar Pradesh	18	148.2	48.1
Madhya Pradesh	14	154.3	50.4
Punjab	11	151.7	49.6
Maharashtra	9	146.8	47.3
Haryana	7	145.2	46.9

1. **Weekday vs Weekend AQI** – Checked whether air quality differs between weekdays (Mon–Fri) and weekends (Sat–Sun).

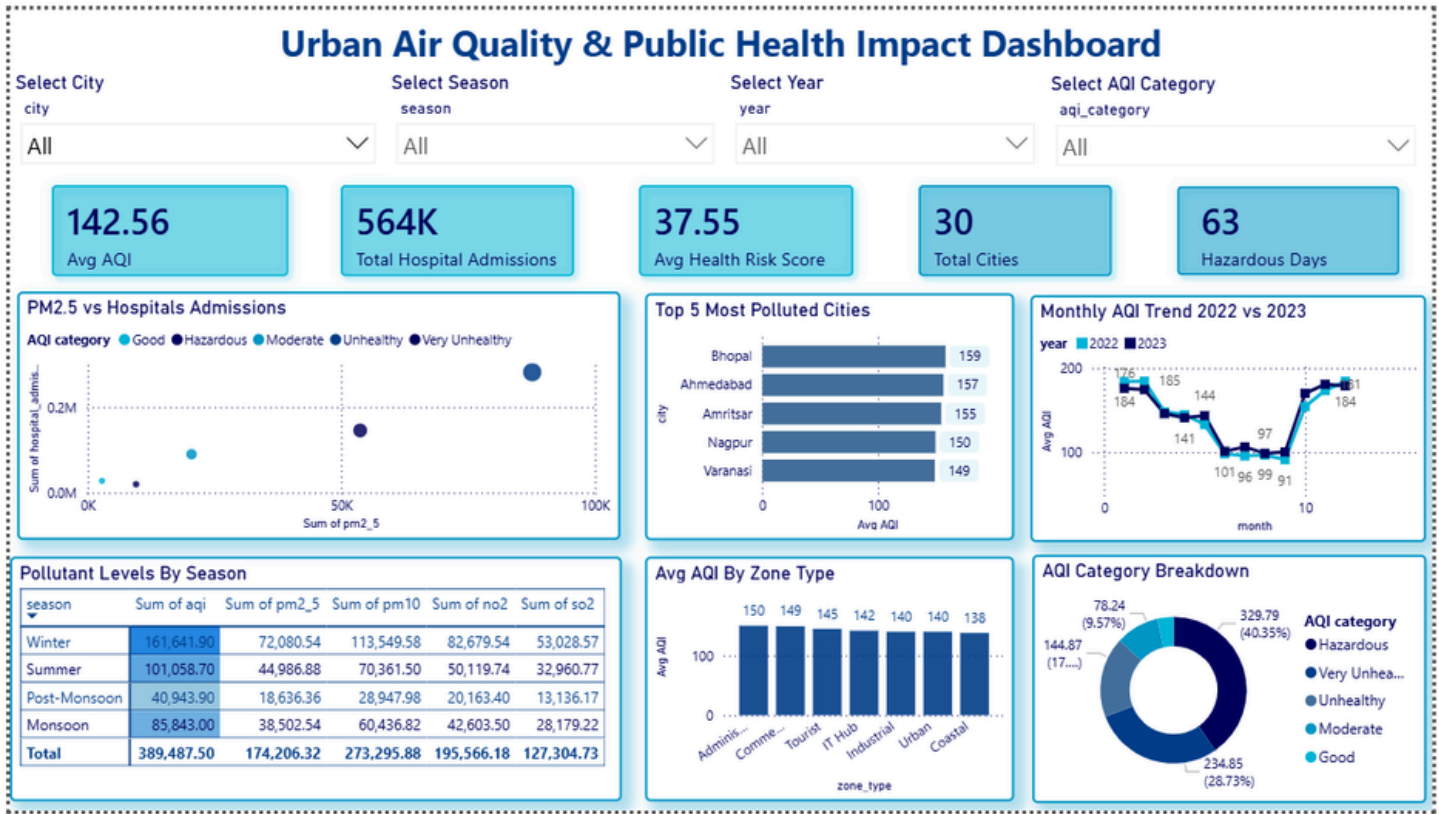
day type	total records	avg aqi	avg pm2_5	avg admissions
Weekday	1,951	144.2	64.8	248
Weekend	781	138.7	61.3	231

```
-- Which are the top 10 most polluted cities?
SELECT
    city,
    state,
    ROUND(AVG(aqi), 2) AS avg_aqi,
    ROUND(AVG(pm2_5), 2) AS avg_pm2_5,
    ROUND(AVG(health_risk_score), 2) AS avg_health_risk,
    COUNT(*) AS total_records
FROM air_quality
GROUP BY city, state
ORDER BY avg_aqi DESC
LIMIT 10;
```

```
-- Which season has the worst air quality?
SELECT
    season,
    ROUND(AVG(aqi), 2) AS avg_aqi,
    ROUND(AVG(pm2_5), 2) AS avg_pm2_5,
    ROUND(AVG(no2), 2) AS avg_no2,
    ROUND(AVG(so2), 2) AS avg_so2,
    ROUND(AVG(hospital_admissions), 2) AS avg_hospital_admissions,
    COUNT(*) AS total_records
FROM air_quality
GROUP BY season
ORDER BY avg_aqi DESC;
```

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



The dashboard includes:

- **KPI Cards:** Avg AQI (142.56), Total Hospital Admissions (564K), Avg Health Risk Score (37.55), Total Cities (30), Hazardous Days (63)
- **Top 10 Most Polluted Cities** — Horizontal bar chart with city-wise average AQI
- **Monthly AQI Trend 2022 vs 2023** — Dual line chart comparing yearly pollution trends
- **AQI Category Breakdown** — Donut chart showing distribution across 5 categories
- **Pollutant Levels by Season** — Matrix heatmap with gradient coloring
- **PM2.5 vs Hospital Admissions** — Scatter plot with trend line and color-coded AQI categories
- **Avg AQI by Zone Type** — Column chart comparing all zone types
- **Slicers:** City, Season, Year, AQI Category for interactive filtering

6. Business Recommendations

- **Target Winter Interventions** – Implement vehicle restriction and industrial regulation policies during Winter months when AQI peaks above 180. Alert systems should be activated from November to February.
- **Focus on Industrial Zones** – Industrial zone cities like Bhopal, Ahmedabad, and Amritsar consistently show highest pollution levels. Enforce stricter emission controls and mandatory green buffer zones.
- **Strengthen Healthcare Preparedness** – Hospital capacity planning should be scaled up during Winter and Post-Monsoon seasons when admissions spike. Respiratory and cardiovascular care units need additional resources.
- **Increase Green Cover** – Cities with higher green cover percentage show lower AQI levels. Target a minimum 25% green cover in all Industrial and Urban zone cities.
- **Promote Public Transport on High-AQI Days** – Weekday pollution is consistently higher than weekends due to vehicle traffic. Subsidized public transport on weekdays can reduce NO₂ and CO emissions significantly.
- **Real-time AQI Monitoring Expansion** – Cities with fewer monitoring stations show data gaps. Increase monitoring stations in Tier-2 cities to improve data accuracy and early warning capabilities.

7. Tools & Technologies Used

Python (Jupyter Notebook) - Data cleaning, EDA, feature engineering, visualization

Pandas & NumPy - Data manipulation and mathematical operations

Matplotlib & Seaborn - Matplotlib & Seaborn

MySQL Workbench - Data storage, SQL queries, business analysis

Power BI Desktop - Interactive dashboard and DAX measures.

GitHub - Version control and project portfolio upload.